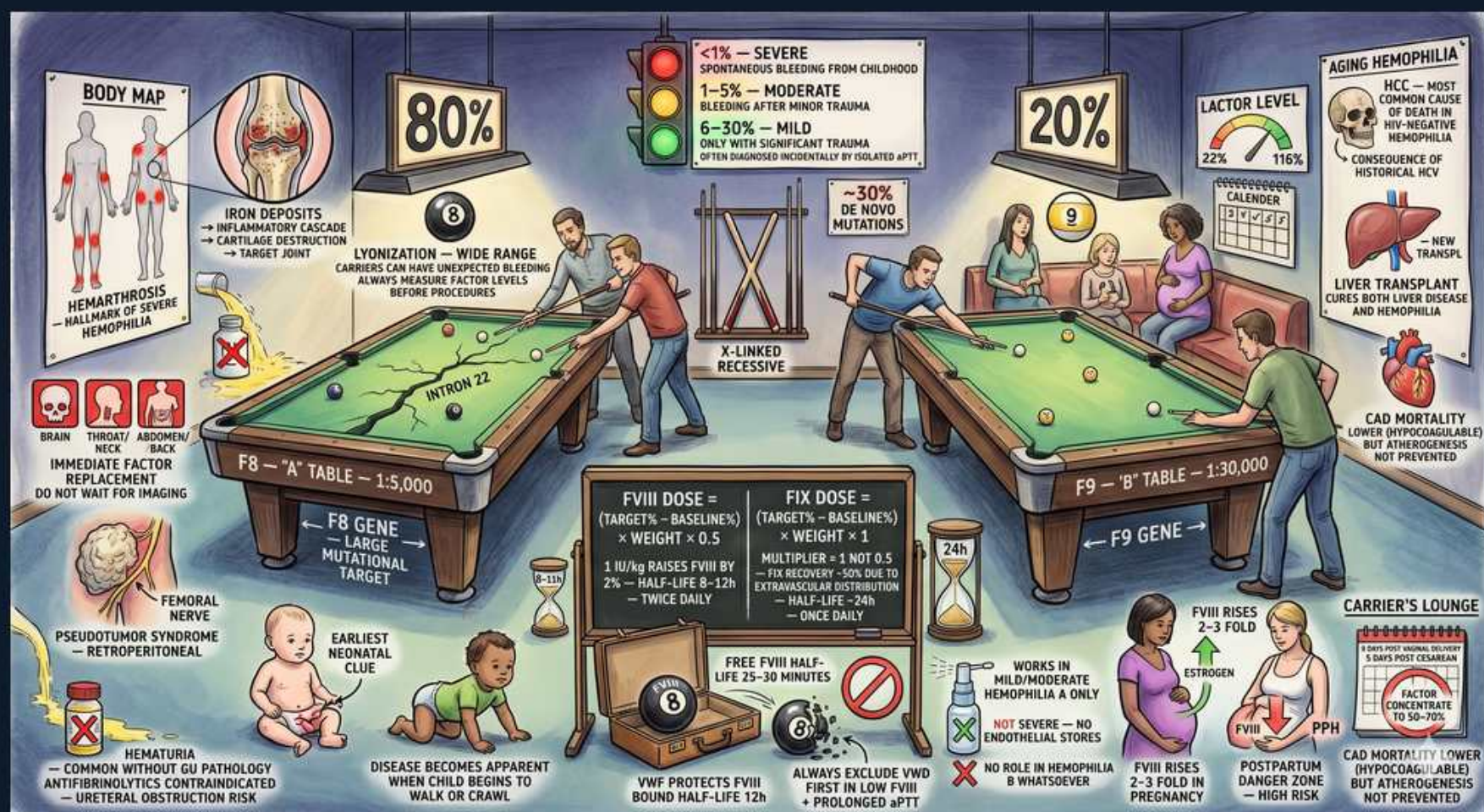


Coagulation — Hemophilia A & B (presentation, labs, dosing)



1. Hemarthrosis — hallmark

WHAT YOU SEE

Body map top-left with red joint and 'hemarthrosis' label

WHAT IT ENCODES

Hemarthrosis — recurrent spontaneous bleeding into large weight-bearing joints (knees, ankles, elbows) — is the hallmark of severe hemophilia. Repeated bleeds cause synovial iron deposition (hemosiderin), chronic synovitis and cartilage destruction, producing a 'target joint' and crippling hemophilic arthropathy. Acute episodes present with a warm, swollen, painful joint held in flexion.

BOARD TRAP

Hemophilia causes DEEP bleeding — hemarthroses, intramuscular hematomas (iliopsoas → femoral nerve palsy, retroperitoneal), and delayed bleeding after trauma/surgery — NOT the immediate mucocutaneous bleeding (epistaxis, petechiae, menorrhagia, gum bleeding) of platelet disorders and von Willebrand disease. Mucocutaneous pattern → think platelet/vWF, not hemophilia.

2. Intracranial bleed danger

WHAT YOU SEE

Brain icon 'immediate factor replacement, do not wait for imaging'

WHAT IT ENCODES

Intracranial hemorrhage is the leading cause of bleeding death in hemophilia and can occur spontaneously or after minor head trauma. The rule: give factor replacement IMMEDIATELY to a level of ~80–100% for any suspected serious or CNS bleed — do this BEFORE and without waiting for CT or neuro consultation.

BOARD TRAP

Do NOT delay factor replacement to obtain imaging or labs in a hemophiliac with suspected ICH or major bleed — 'treat first, image second' is the tested principle; ordering the CT before giving factor is the wrong answer.

3. Factor VIII = Hemophilia A

WHAT YOU SEE

The big left 'A' pool table labelled 1:5,000 / F8 — Hemophilia A = factor VIII, the common one

WHAT IT ENCODES

Hemophilia A = factor VIII deficiency, the most common severe inherited bleeding disorder, ~1 in 5,000 male births and ~80–85% of hemophilia cases. It is X-linked recessive, so it overwhelmingly affects males. FVIII is part of the intrinsic pathway, so its deficiency prolongs the aPTT.

BOARD TRAP

Hemophilia A = factor VIII (the more common 'A'), NOT factor IX; mixing up A=VIII vs B=IX is the most basic and most tested distinction — mnemonic: A comes first alphabetically and is the more common/8.

4. Factor IX = Hemophilia B

WHAT YOU SEE

Right 'F9 B table' factor IX deficiency 1:30,000

WHAT IT ENCODES

Hemophilia B (Christmas disease) = factor IX deficiency, ~1 in 30,000 males, also X-linked recessive. It is clinically INDISTINGUISHABLE from hemophilia A — same hemarthrosis, same deep bleeds, same prolonged aPTT with normal PT — so only a specific factor assay can tell them apart.

BOARD TRAP

Hemophilia B = factor IX (Christmas disease), not VIII; you CANNOT distinguish A from B by history, exam or routine coags — the board answer when asked to differentiate is 'specific factor VIII and IX activity assays.' Also note DDAVP does not work in B.

5. X-linked recessive

WHAT YOU SEE

A central 'X-LINKED RECESSIVE' sign with the giant X between the two tables — sex-linked inheritance affecting males

WHAT IT ENCODES

Both hemophilia A and B are X-linked recessive. Affected fathers cannot pass it to sons (sons get the Y) but all daughters become obligate carriers; carrier mothers pass it to ~50% of sons (affected) and ~50% of daughters (carriers). Rarely, carrier females bleed due to lyonization (skewed X-inactivation).

BOARD TRAP

Hemophilia is X-LINKED RECESSIVE, not autosomal — the absence of male-to-male transmission and presence of affected maternal uncles is the pedigree giveaway; females are usually asymptomatic carriers but can be symptomatic with extreme lyonization.

6. Prolonged aPTT, normal PT

WHAT YOU SEE

A center clue under the table 'normal PT, prolonged aPTT' — the isolated intrinsic-pathway lab signature

WHAT IT ENCODES

The lab signature of hemophilia is an isolated prolonged aPTT with a NORMAL PT, normal platelet count, and normal bleeding time/PFA. A mixing study CORRECTS (1:1 mix with normal plasma normalizes the aPTT), confirming a factor DEFICIENCY rather than an inhibitor. PT stays normal because FVIII/FIX are intrinsic-pathway factors.

BOARD TRAP

PT is NORMAL in hemophilia — an abnormal PT points elsewhere (DIC, liver disease, vitamin K deficiency, warfarin); and if the mixing study does NOT correct, suspect an inhibitor (acquired hemophilia or alloantibody), not simple deficiency.

7. Severity by factor level

WHAT YOU SEE

Stoplight severity gauge top-center (<1% severe, 1-5% mod, 6-30+% mild)

WHAT IT ENCODES

Severity is graded by residual factor activity: <1% = severe (spontaneous hemarthroses, frequent bleeds), 1–5% = moderate (bleeds with minor trauma), >5–40% = mild (bleeds only with significant trauma or surgery). This is why severe disease presents in infancy and mild disease may go unrecognized for decades.

BOARD TRAP

Mild hemophilia (>5% activity) can have an aPTT that looks nearly normal and may first present only at surgery, dental extraction or major trauma — a normal-appearing screening aPTT does not exclude mild hemophilia; a strong bleeding history warrants factor assays.

8. ~30% de novo mutations

WHAT YOU SEE

Right '~30% de novo mutations' note

WHAT IT ENCODES

About one-third of hemophilia cases (~30%) arise from NEW (de novo) spontaneous mutations with no prior family history. The most common severe FVIII mutation is the intron 22 inversion. Thus a negative family history is common and does not argue against the diagnosis.

BOARD TRAP

Absence of a family history does NOT rule out hemophilia — roughly 30% are de novo mutations; do not dismiss a male infant with hemarthrosis/circumcision bleeding because 'no one in the family has it.'

9. vWF protects FVIII

WHAT YOU SEE

A center note 'vWF protects FVIII — free FVIII half-life ~12h' — the carrier protein that keeps FVIII from being cleared

WHAT IT ENCODES

Von Willebrand factor is the plasma carrier that binds and stabilizes FVIII; free (unbound) FVIII has a half-life of only ~12 hours, but bound to vWF it is protected from clearance. Consequently, in von Willebrand disease (especially type 2N and severe type 3), FVIII is also low because its chaperone is missing.

BOARD TRAP

Always EXCLUDE von Willebrand disease before labeling an isolated low FVIII as hemophilia A — vWD is far more common, can lower FVIII, and (unlike hemophilia) shows mucocutaneous bleeding and abnormal vWF/ristocetin testing; type 2N vWD in particular masquerades as mild hemophilia A.

10. FVIII dosing rule

WHAT YOU SEE

An hourglass-flanked formula 'FVIII dose = (target - baseline) x weight x 0.5' — 1 IU/kg raises FVIII ~2%

WHAT IT ENCODES

FVIII dosing: dose (IU) $\approx$ weight(kg) $\times$ 0.5 $\times$ (target% - baseline%), because 1 IU/kg of FVIII raises the plasma level by ~2%. FVIII half-life is ~8–12h, so it is dosed roughly twice daily for ongoing bleeds. Targets: ~30–50% for minor bleeds, ~50–80% for major, ~80–100% for life-threatening/CNS bleeds or major surgery.

BOARD TRAP

The multipliers differ between factors — FVIII uses the $\times 0.5$ factor (1 IU/kg $\rightarrow$ ~2% rise); applying the FIX dosing constant to FVIII (or vice versa) under- or over-doses the patient. Know that FVIII raises level twice as efficiently per IU/kg as FIX.

11. FIX raises ~1% per IU/kg

WHAT YOU SEE

The right F9-table dosing 'multiplier $\times 1$ — 1 IU/kg raises FIX ~1%' — FIX needs double the units per kg

WHAT IT ENCODES

FIX dosing: dose (IU) $\approx$ weight(kg) $\times$ 1 $\times$ (target% - baseline%), because 1 IU/kg of FIX raises the level only ~1% — FIX has a larger volume of distribution (binds collagen/endothelium) so you need roughly DOUBLE the units per kg compared with FVIII. FIX half-life is longer (~18–24h), allowing once-daily dosing.

BOARD TRAP

Do NOT use the FVIII multiplier for factor IX — FIX needs about twice the dose per kg for the same percentage rise because of its larger distribution volume; this dosing-constant swap is a favorite calculation trap.

12. Desmopressin (DDAVP) mild A

WHAT YOU SEE

A right-side DDAVP note 'works in mild/moderate hemophilia A only — no role in B' — releases stored FVIII/vWF

WHAT IT ENCODES

Desmopressin (DDAVP) triggers release of stored FVIII and vWF from endothelial Weibel-Palade bodies, transiently raising FVIII ~2–4 fold — useful for MILD (and some moderate) hemophilia A and type 1 vWD. It avoids blood-product exposure. Tachyphylaxis develops after repeated doses (stores deplete), and watch for hyponatremia/fluid retention.

BOARD TRAP

DDAVP works ONLY in mild/moderate hemophilia A — it does NOTHING in hemophilia B (no FIX to release) and is ineffective in SEVERE hemophilia A (there is essentially no FVIII to release); offering DDAVP for a severe-A or any-B bleed is wrong.

13. Carrier / pregnancy FVIII rises

WHAT YOU SEE

The 'carrier's lounge' pregnant figure 'FVIII rises 2-3 fold, postpartum danger' — estrogen masks then unmasks bleeding

WHAT IT ENCODES

Pregnancy physiologically raises FVIII and vWF 2–3 fold (estrogen effect), which can normalize the FVIII of a mild-hemophilia-A carrier and even an affected woman during gestation, providing hemostatic cover for delivery. After delivery these levels fall rapidly, creating a postpartum/secondary hemorrhage danger window.

BOARD TRAP

Normal FVIII levels during pregnancy do NOT exclude carrier status or mild hemophilia A — measure at baseline (non-pregnant) and remember the postpartum drop predisposes to delayed bleeding; reassure-and-discharge based on a third-trimester FVIII is a trap.